

A PROJECT TITLE
ON
Feature Importance Analysis for Player Performance Using Random Forest and XGBoost
project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY**

in
A.Y.: 2026-27

by

Name: M. Jagadish roll number : 2520030165

Name: M. Yashwant Raj roll number : 2520030181

Under the Esteemed guidance of

Mr. Shaik Asif

Assistant Professor

CSE
COMPUTER SCIENCE AND ENGINEERING



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in





Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Feature Importance Analysis for Player Performance Using Random Forest and XGBoost

1. Abstract:

Player performance analysis has become an essential aspect of modern sports analytics, helping coaches, analysts, and team managers make informed decisions based on data rather than intuition. This project presents a machine learning approach to identify the most influential factors affecting player performance using two powerful ensemble learning algorithms: **Random Forest** and **XGBoost**. The primary objective is to determine which performance metrics contribute most significantly to predicting a player's overall effectiveness, thereby enabling better talent evaluation, team selection, and strategic planning.

The system begins by collecting and preprocessing player performance data, including features such as matches played, goals scored, assists, passing accuracy, tackles, interceptions, fitness levels, and other relevant statistics. Data preprocessing techniques such as handling missing values, encoding categorical variables, feature scaling, and outlier detection are applied to improve data quality and model performance. The processed dataset is then divided into training and testing subsets for model development and evaluation.

Random Forest is employed to generate feature importance scores based on the reduction in impurity across multiple decision trees, while XGBoost utilizes gradient boosting techniques to rank features according to their contribution to prediction accuracy. The performance of both models is evaluated using metrics such as accuracy, precision, recall, F1-score, and Mean Squared Error (for regression-based predictions). Comparative analysis is performed to assess the consistency and reliability of feature rankings produced by both algorithms.

The proposed system provides clear visualizations of feature importance through bar charts and ranking plots, enabling stakeholders to understand which attributes have the greatest impact on player performance. This insight supports evidence-based decision-making in player scouting, training optimization, injury prevention, and match strategy formulation. By combining advanced machine learning techniques with explainable feature importance analysis, the project demonstrates an efficient and practical framework for sports performance evaluation, offering valuable insights for coaches, analysts, sports organizations, and researchers seeking to leverage data-driven methodologies in competitive sports.

2. INTRODUCTION:

In today's sports industry, data analytics plays a vital role in evaluating player performance and supporting strategic decision-making. Traditional methods of assessing players often rely on subjective opinions and manual observation, which may overlook hidden patterns in performance data. Machine learning provides an efficient solution by analyzing large volumes of player statistics to identify the factors that significantly influence performance.

This project, "**Feature Importance Analysis for Player Performance Using Random Forest and XGBoost**," aims to determine the most influential features affecting a player's overall performance. The system uses historical player statistics such as matches played, goals, assists, passing accuracy, tackles, interceptions, shots on target, fitness level, and other relevant attributes. After preprocessing the dataset, two advanced ensemble machine learning algorithms—**Random Forest** and **XGBoost**—are trained to predict player performance and rank the importance of each feature. **Random Forest** identifies feature importance by measuring how much each feature reduces impurity across multiple decision trees, while **XGBoost** calculates feature importance based on each feature's contribution to improving prediction accuracy during the boosting process. Comparing the results from both models provides a more reliable understanding of the factors influencing player performance. The proposed system assists coaches, sports analysts, and team managers in making data-driven decisions related to player selection, performance improvement, scouting, and match strategy. Additionally, the visualization of feature importance helps users easily interpret the results, making the system both accurate and explainable. Overall, this project demonstrates how machine learning can transform sports analytics by providing objective insights into player performance.

3. SOFTWARE REQUIREMENTS:

Operating System

- Windows 10/11, Linux, or macOS

Programming Language

- Python 3.9 or above

Development Environment

- Jupyter Notebook
- Google Colab
- Visual Studio Code (VS Code)

Libraries and Frameworks

- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Matplotlib
- Seaborn
- Joblib (for model saving)

Dataset

- Player Performance Dataset (CSV format)

Machine Learning Algorithms

- Random Forest Regressor/Classifier
- XGBoost Regressor/Classifier

Visualization Tools

- Matplotlib
- Seaborn

Hardware Requirements

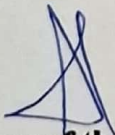
- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** 10 GB free disk space
- **Internet:** Required for downloading datasets and Python packages

Software Dependencies

- Python 3.x
- Anaconda (Optional)
- pip Package Manager
- Git (Optional for version control)

Performance Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R^2 Score (for regression)
- Feature Importance Scores (Random Forest & XGBoost)



Signature of the Guide



Course Instructor